



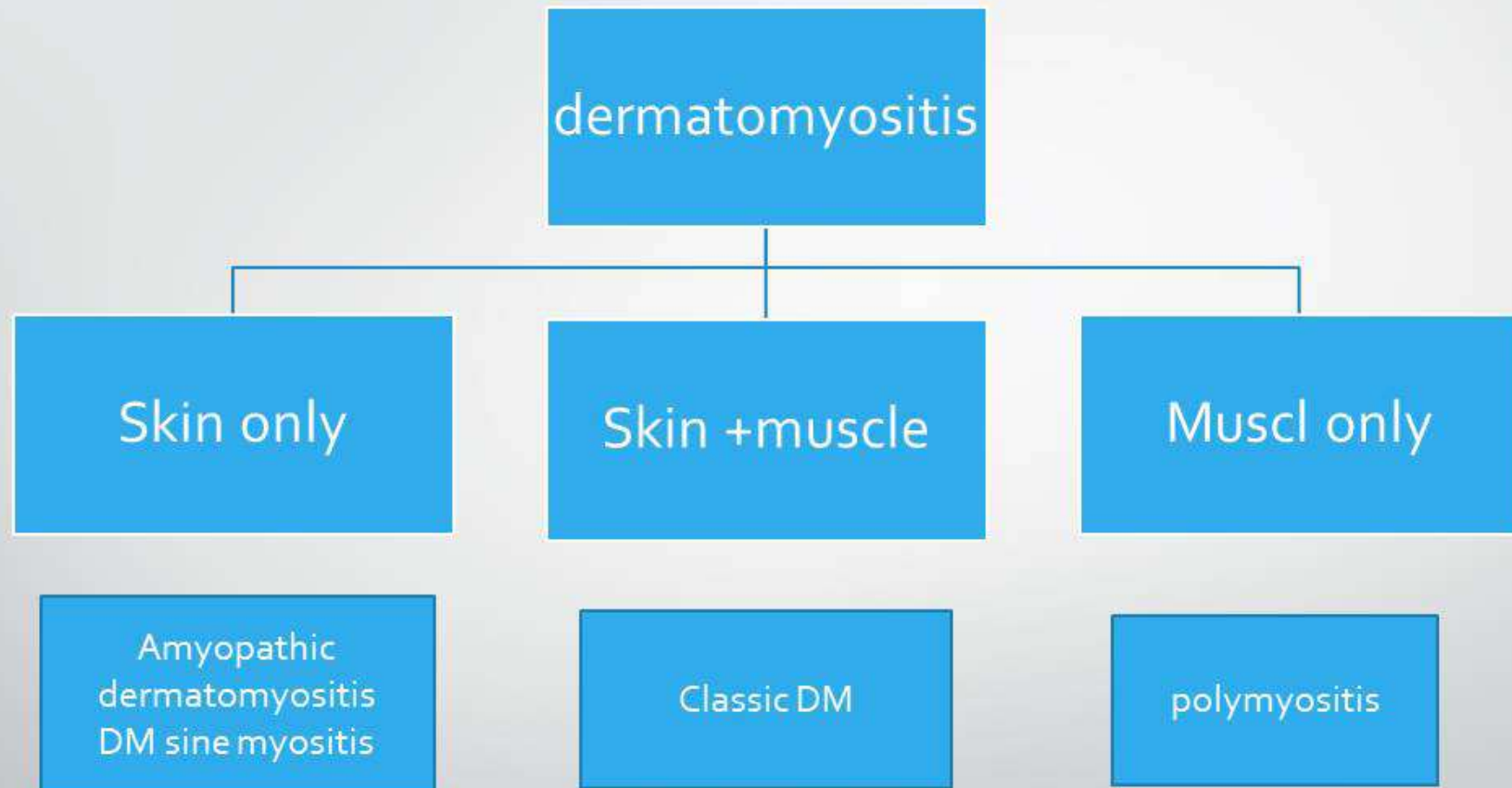
Other connective tissue disease

Dr eman gomaa
Consultant dermatology

Dermatomyositis

- Multisystemic autoimmune disease affecting muscle, skin, other
- Rare disease, world wide,
- Bimodal :
 - Juvenile
 - Adult : women 3:1









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Aetiopathogenesis

Enviromental
triggers

In
Genaticly
susptable pt

Immune
dysregulation

Triggers

1. Malignancy
2. Infection:
3. Drugs

Genetic

Monozygotic twins
HLA : DR3 , DR 52, DR7

IMMUNOLOGICAL

1. Anti JO1
2. Anti Ml2

Infection

:picorna viruse , echoviruse
,coxackie viruse
AIDS , E coli

Drugs

NSAID, statin , hydroxyurea , d
penicillamine , tnf alpha ,
cyclophosphamide



Skin

Systemic

Malignancy association

Skin manifestation



Fig. 42.3 Dermatomyositis – marked facial involvement. Note the violaceous hue. Misdiagnoses include psoriasis, contact dermatitis, and if there is significant facial involvement as in this patient, acute cutaneous lupus erythematosus.

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- Poikiloderma (sever itchy)
 - Pink _ violet
 - Heliotrope erythema
 - Shawl sign
 - V sign
- Gottron sign
- Gottron papules
- Nail fold changes
 - Ragged cuticle
 - Telangectasia
- Calcinosis cutis
- Csvv
- Overlab syndrome , mixed CT disease



Fig. 42.5 Dermatomyositis – Gottron sign. Pink-violet papules (A) and thin pink plaques (B) of the elbows and knee (C). Some of the papules on the elbow are flat-topped (lichenoid) and others have white scale. The presence of plaques on the elbows and knees can lead to the misdiagnosis of psoriasis. *BC, Courtesy, Julie V Schaffer, MD.*



Fig. 42.4 Dermatomyositis – Gottron papules. A Continuation of the pink-violet inflammation over the metacarpophalangeal (MCP) joints and proximal and distal interphalangeal (PIP, DIP) joints as well as the proximal nail folds. Note the ragged cuticles. B The flat-topped (lichenoid) papules overlying the DIP, PIP, and MCP joints (knuckles) are subtle and were misdiagnosed as verrucae vulgares in this child. C More obvious disease in which the multiple pink-violet lichenoid papules are coalescing and there is some involvement of the interphalangeal skin. *A, Courtesy, Kalman Watsky, MD; B, Courtesy, Julie V Schaffer, MD.*



Fig. 42.6 Dermatomyositis – involvement of the upper back. The pink-violet plaques, some with associated scale, were very pruritic, as evidenced by multiple excoriations. Linear streaks of erythema are also seen. *Courtesy, Jean I. Rabinovici, MD.*



Fig. 42.8 Juvenile dermatomyositis – calcinosis cutis. The nodules favor the extensor surface of the knees and result in a decrease in joint mobility.



Fig. 42.9 Dermatomyositis – less common presentations. A Flagellate erythema of the posterior trunk. B Cutaneous erosions and ulcerations can

DD



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DIFFERENTIAL DIAGNOSIS OF DERMATOMYOSITIS

Systemic lupus erythematosus*

Physician might notice the nail-fold telangiectasias and photodistributed poikiloderma but miss the muscle weakness, heliotrope, extensor distribution, pruritus, and the violaceous hue (Fig. 42.3)
Most commonly an issue when patient has amyopathic dermatomyositis and presents with a photosensitive eruption and no muscle weakness

Psoriasis

Involvement of elbows and knees with papulosquamous lesions can lead to misdiagnosis. In addition, Gottron papules, extensor surfaces, and scalp can have psoriasiform scaling

Airborne or allergic contact dermatitis

Eyelid edema and lichenification secondary to pruritus can be marked in dermatomyositis; look for additional sites of dermatitis

Photodrug eruption

Photodistribution

Cutaneous T-cell lymphoma

The poikiloderma often begins in intertriginous zones rather than on the scalp, face, and extensor surfaces

Atopic dermatitis

Usually in children, where the physician focuses on the pruritus and secondary lichenification

Systemic sclerosis*

The nail-fold telangiectasias may be similar in appearance, but the dyspigmentation is quite different; edema of the hands is an early sign of systemic sclerosis

Trichinosis

Patients have painful muscles and periorbital edema, but not other features

Photodistributed form of multicentric reticulohistiocytosis

Firm papules have distinct histologic features and patients have no nail-fold changes associated with dermatomyositis

Systemic affection

1. Constitutional

2. Myopathy

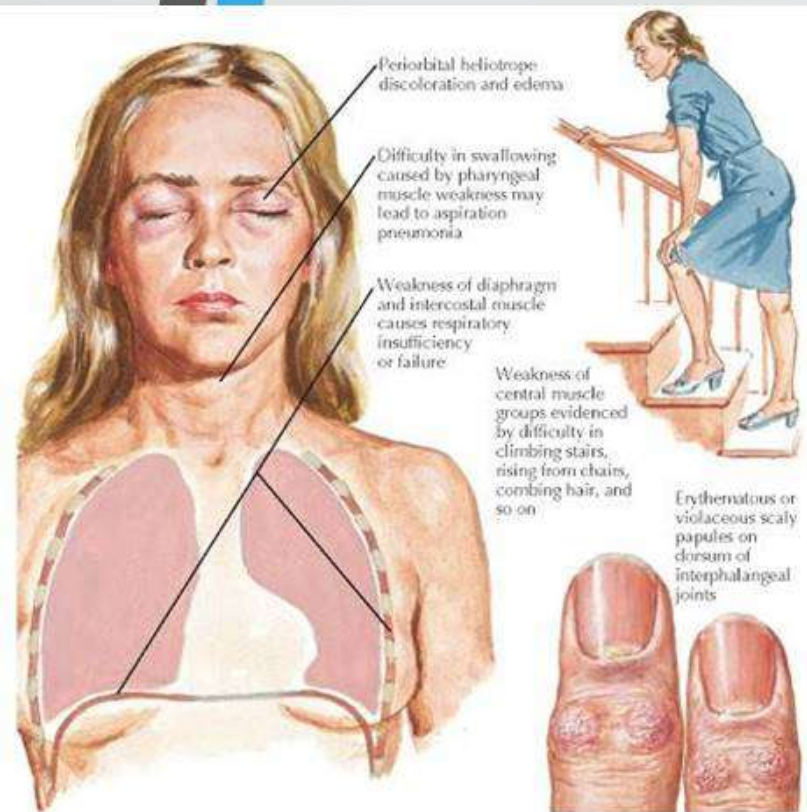
Proximal , extensor ,
bilateral , symmetrical ,
Tender on palpation
Unable to comb hair , up stair

3. GIT :GERD

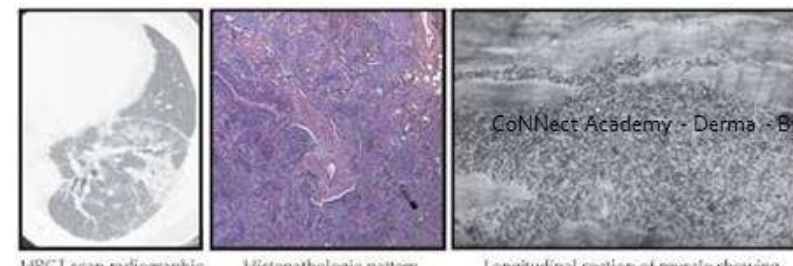
dysphagia :cricopharyngeal muscle

4. Cardiac : arrhythmia

5. Pulmonary :diffuse interstitial fibrosis



Erythematous or violaceous scaly papules on dorsum of interphalangeal joints



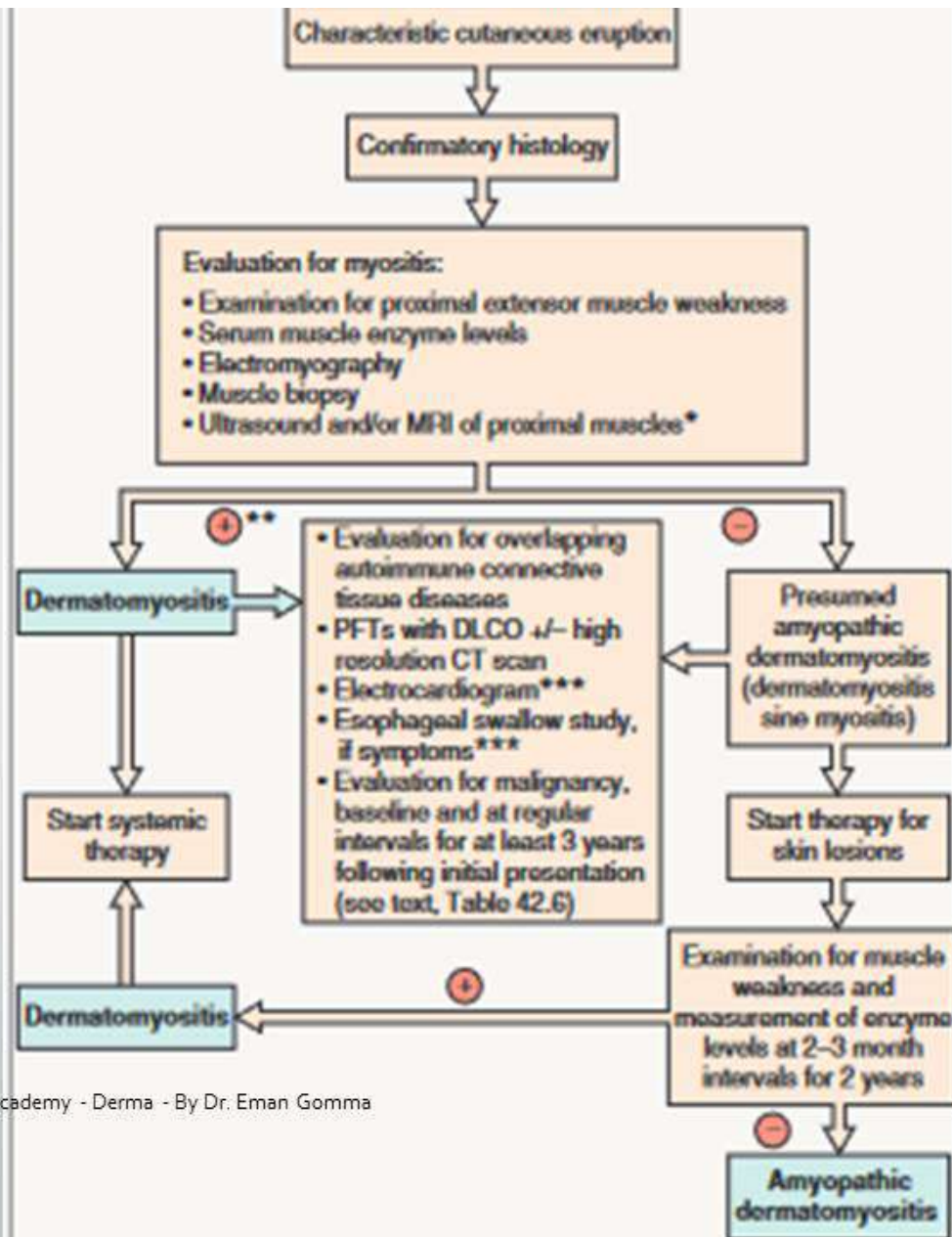
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HRC1 scan radiograph Histopathologic pattern Longitudinal section of muscle showing

Association

- Malignancy
 - Adult DM , amyopathic DM
 - Ovarian , colon , breast
 - Not in juvenile DM , polymyositis
- Other AICTD
- Immunological disease

Causes of death in DM: respiratory infection, cardiac failure, malnutrition (difficulty in swallowing), carcinoma,



Investigation

Skin biopsy

Serum

Ana, anti JO1, anti
Ml2
CPK
Aldolase
LDH, myoglobin, ast,
alt

Muscle
MRI
Ultrasound
electromyography
biopsy

Urine
Creatinine
Myoglobine

Other to
detect
malignancy

Biopsy of muscle :triceps

Lymphocytic infiltrate perifascicular,
perivascular, muscle atrophy,
necrosis.

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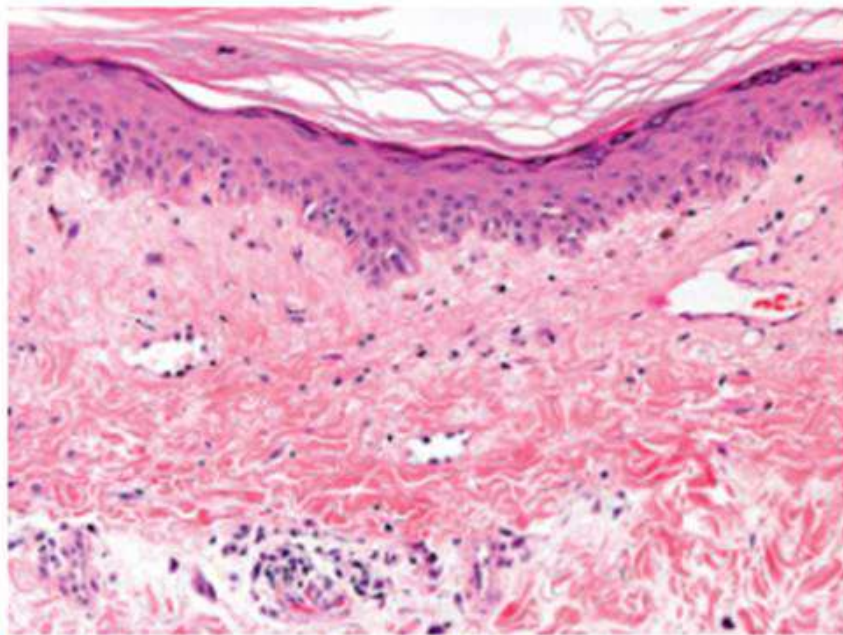


Fig. 42.10 Dermatomyositis – histopathologic features. The vacuolar changes within the basal layer can be subtle. In the dermis, dilated blood vessels and a sparse lymphocytic infiltrate are seen (poikilodermatous changes). Courtesy, Lorenzo Cerami, MD.

-) **Skin biopsy:**
 - Poikilodermatous lesions: LE-like changes
 - Epidermal changes: atrophy, basement membrane degeneration, vacuolar alteration of basal keratinocytes.
- - Dermal changes: interstitial mucin deposition, V.D and a sparse lymphocytic infiltrate.
- - Gottron's papules (more specific): lichenoid infiltrate + acanthosis (NOT epidermal atrophy)

Treatment

Skin

Challenging not all patient
clear

1. Photoprotection
2. Antimalarial
3. Low dose methotrexate
4. Other immunosuppressive

العضلات هتتحسن
الجلد مش هيتحسن!!!

Muscle

Systemic corticosteroid
Steroid sparing
Rest, physiotherapy

THERAPEUTIC LADDER FOR DERMATOMYOSITIS



Systemic therapy

Oral prednisone: 1 mg/kg/day tapered to 50% over 6 months and to zero over 2–3 years (1)

Option to use pulse, split-dose, or alternate-day (2)

Methotrexate: 5–25 mg weekly po, sc, IM (2)

Azathioprine: 2–3 mg/kg/day po (1)

Others:

High-dose IVIg (2 g/kg/month) (1)

Mycophenolate mofetil (1 g po twice daily) (2)

Pulse cyclophosphamide (0.5–1.0 g/m² IV monthly) (2)

Chlorambucil (4 mg/day po) (2)

Cyclosporine (3–5 mg/kg/day po) (2)

Rituximab (375 mg/m²/infusion for 4 weekly IV infusions) (1)

Tacrolimus (0.12 mg/kg/day po) (3)

Sirolimus (5 mg/day × 2 weeks, 2 mg/day × 2 weeks, then 1 mg/day po) (3)

TNF- α inhibitors (e.g. infliximab, etanercept) (3)*

Fludarabine (3)

Hematopoietic stem cell transplantation (3)

Plasmapheresis (3)[†]

Cutaneous lesions

Photoprotection including sunscreens (high SPF including protection against UVA) and clothing (3)

Topical corticosteroids (3)

Topical tacrolimus (3)

Hydroxychloroquine (200 mg po twice daily; increased frequency of drug eruptions in patients with dermatomyositis) or chloroquine (250 mg po daily) (2)

Hydroxychloroquine (200 mg twice daily) or chloroquine (250 mg po daily) plus quinacrine (100 mg po daily) (3)

Low-dose weekly methotrexate (5–15 mg weekly po, sc, IM) (2)

Mycophenolate mofetil (3)

High-dose IVIg (2 g/kg/month) (1)

Retinoids (3)

Dapsone (3)

Thalidomide (3)

Lenalidomide (3)

Antiestrogens (e.g. tamoxifen, anastrozole) (3)

TNF- α inhibitors (e.g. infliximab, etanercept) (3)*

Rituximab (3)

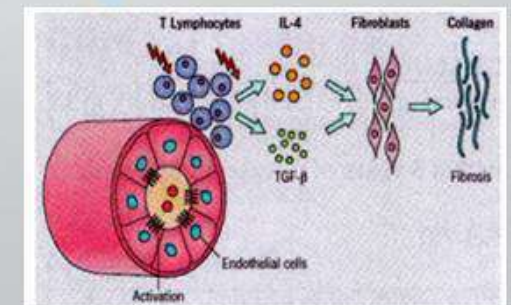
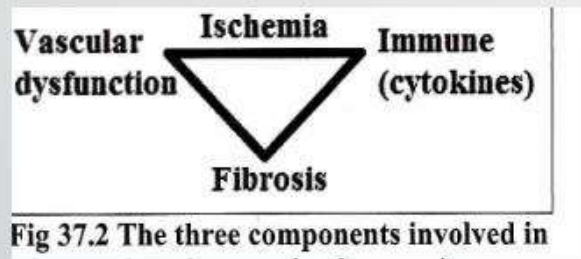
Oral tacrolimus (3)

Systemic sclerosis scleroderma



- Systemic sclerosis (SSc) is a systemic AI-CTD that affects the skin, blood vessels and internal organs.
 1. Skin !=> symmetric hardening of the skin.
 2. Blood vessels Raynaud's phenomenon and digital ulcers.
 3. Internal organ involvement affects the lungs (leading cause of death), gastrointestinal tract, heart, kidneys and other organs.

aetiopathogenesis



Vascular
Abnormal response of BV muscle
endothelial damage >> intimal proliferation >> narrowing B>>
ischemia >> hypoxia

Tissue fibrosis
Hypoxia >> t cell >> il4, TGFB, CTGF >> increase collagen, proteoglycan

Immune dysregulation
Anti centromere
Antitopoisomerase 1

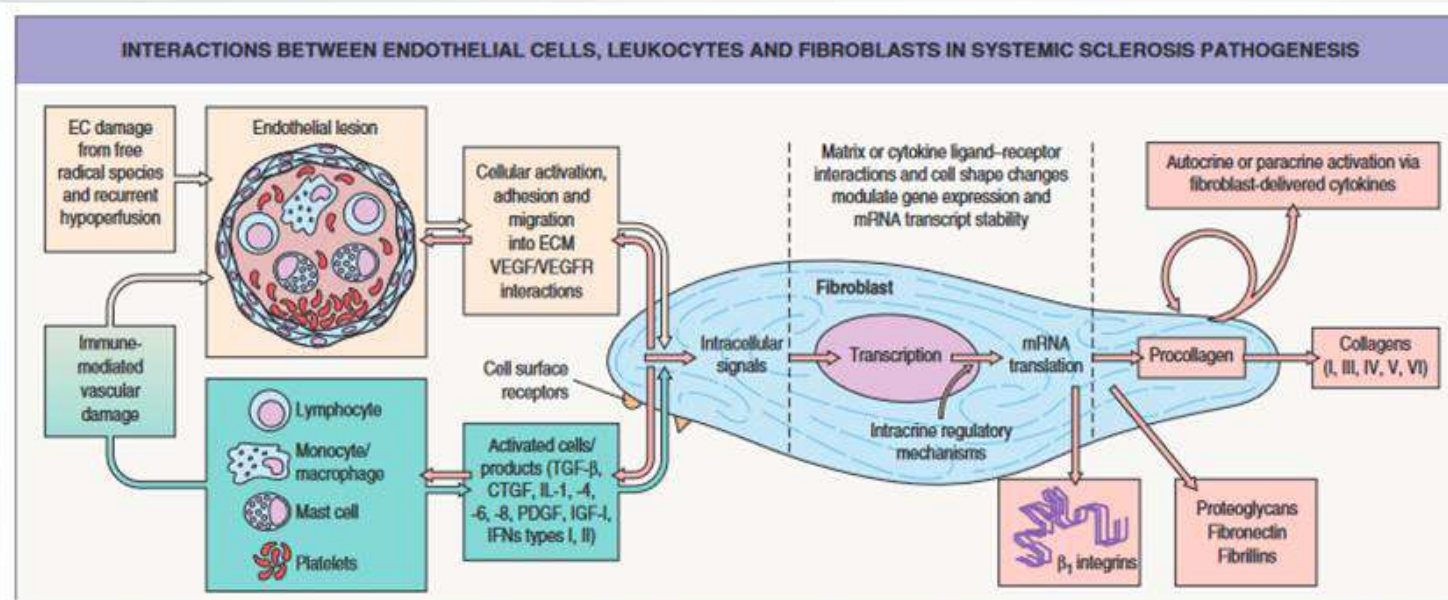
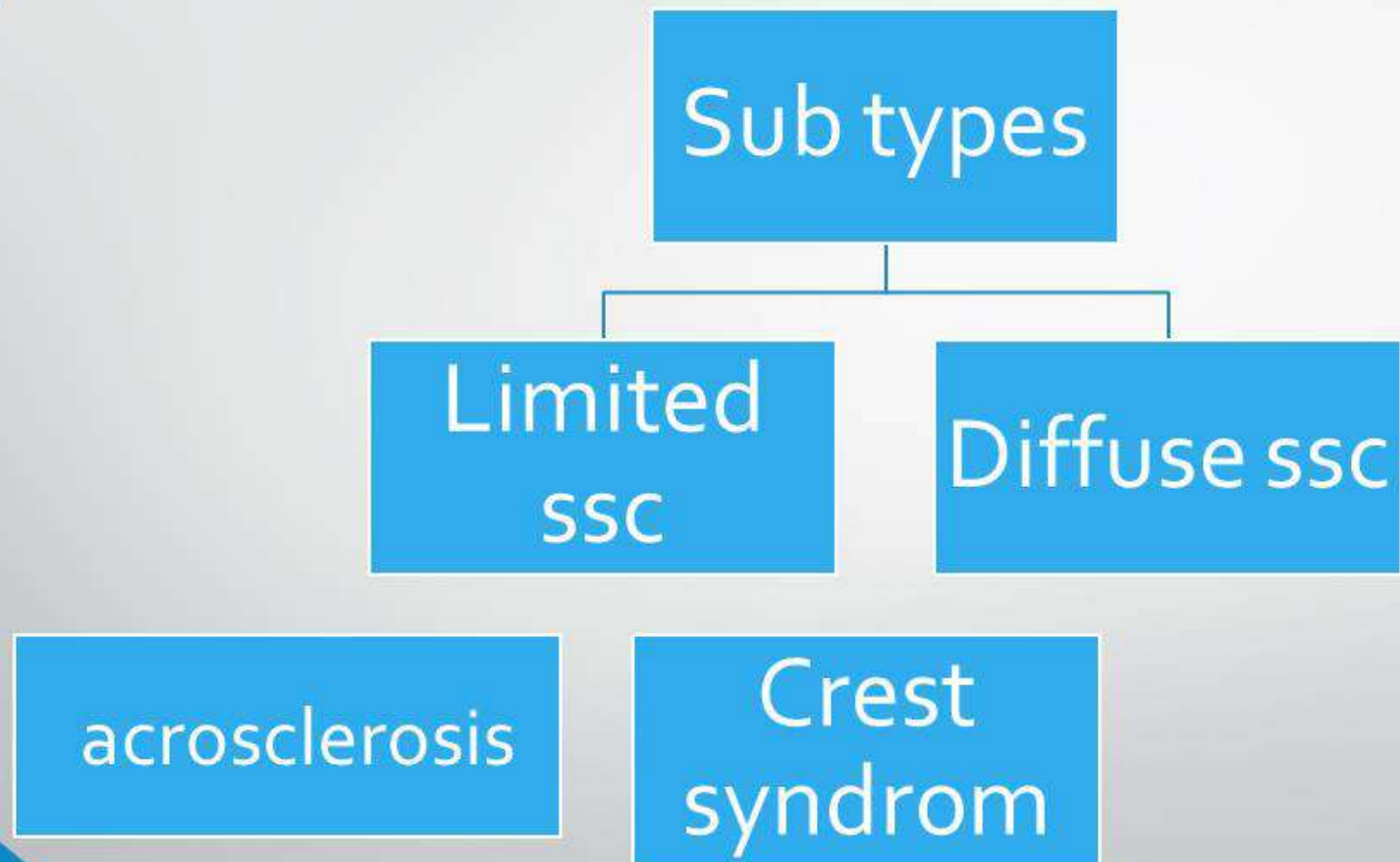


Fig. 43.1 Interactions between endothelial cells, leukocytes and fibroblasts in systemic sclerosis pathogenesis. Genetic susceptibility loci that may increase the risk of developing systemic sclerosis include a region on chromosome 15q (which contains the fibrillin-1 gene) as well as polymorphisms in *STAT4* and the promoter for CTGF. CTGF, connective tissue growth factor; EC, endothelial cell; ECM, extracellular matrix; IFN, interferon; IGF, insulin-like growth factor; PDGF,



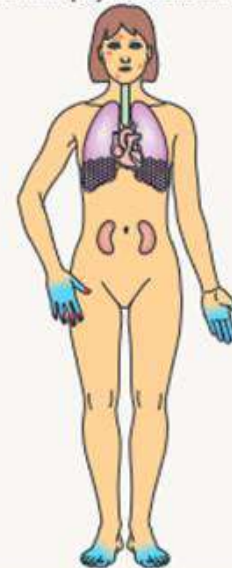
CLINICAL CLASSIFICATION OF SYSTEMIC SCLEROSIS (SSc)

Limited cutaneous SSc (lcSSc) (Anti-centromere antibody)



- Formerly called CREST syndrome
- Long preceding history of Raynaud's phenomenon
- Slower development of limited cutaneous sclerosis involving the distal extremities and face
- Later onset of internal organ involvement (after 10-15 years)
- PAH more common/ILD less common than in diffuse
- More favorable long-term prognosis

Diffuse cutaneous SSc (dcSSc) (Anti-topoisomerase-I antibody) (Anti-RNA polymerase III antibody)



- Sudden onset of Raynaud's phenomenon
- Rapidly progressive, more widespread cutaneous sclerosis (usually peaks within 12-18 months)
- >90% demonstrate internal organ involvement within the first 5 years
- ILD > PAH
- Kidney disease

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- Cutaneous sclerosis
- Raynaud's phenomenon
- Nail fold capillary abnormalities

- Esophageal dysmotility
- Interstitial lung disease (ILD)



- Cardiomyopathy, heart failure
- Pulmonary arterial hypertension (PAH)

- Hypertension renal crisis

The limited symptoms of scleroderma are referred to as **CREST**

Calcinosis- calcium deposits in the skin



Raynaud's phenomenon-
spasm of blood vessels in
response to cold or stress



Esophageal dysfunction- acid reflux and
decrease in motility of esophagus



Sclerodactyly- thickening and tightening
of the skin on the fingers and hands



Telangiectasias- dilation of
capillaries causing red marks
on surface of skin



 **ADAM**

Clinical features cutaneuse

fibrosis	Oedematous phase >>induration >>>atrophic phase (youthful look), peaked nose ,fish mouth ,flexion , contraction .
raynauds	Episodic vasospasm of digital artery ,2 types 1ry , 2ry
ulcer	Tip of digits , inter phalangeal , >osteomyelitis >>amputation
Dyspigmentation	leukoderma >> with retention of perifollicular pigmentation >> salt and pepper Face , upper trunk
telangiectasia	Matted , squared off (face , lips , palm)
Calcinosis cutis	Extremities near joint



Fig. 43.3 Early, edematous phase of systemic sclerosis. Note the demonstration of pitting edema on two of the digits. (Jurel, Ivan / Rokyma, M)





Fig. 43.9 Raynaud phenomenon in a patient with early systemic sclerosis. There is obvious blanching of two fingers as well as violaceous hue of forefinger. *Courtesy, Kelly P. Cohen, MD*



Fig. 43.11 Cutaneous ulcerations in systemic sclerosis. Ulcers favor the digits (A) and sites of trauma such as the elbow (B). Note the calcinosis cuts at the base of the forefinger and the pterygium inversum unguis of the third finger. *A, Courtesy, Nathan Walsh, MD; B, Courtesy, Joyce Ricci, MD.*



Fig. 43.6 The "salt and pepper" sign. Leukoderma with retention of perifollicular pigmentation in a patient with systemic sclerosis. *Courtesy, Joyce Ricci, MD.*

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Fig. 43.7 Mottled (scurvy-like) telangiectasia in systemic sclerosis. The first patient (A) limited cutaneous, while the second patient (B) presented with hyperpigmentation and interstitial lung disease. *Courtesy, Jean L. Rook, MD.*

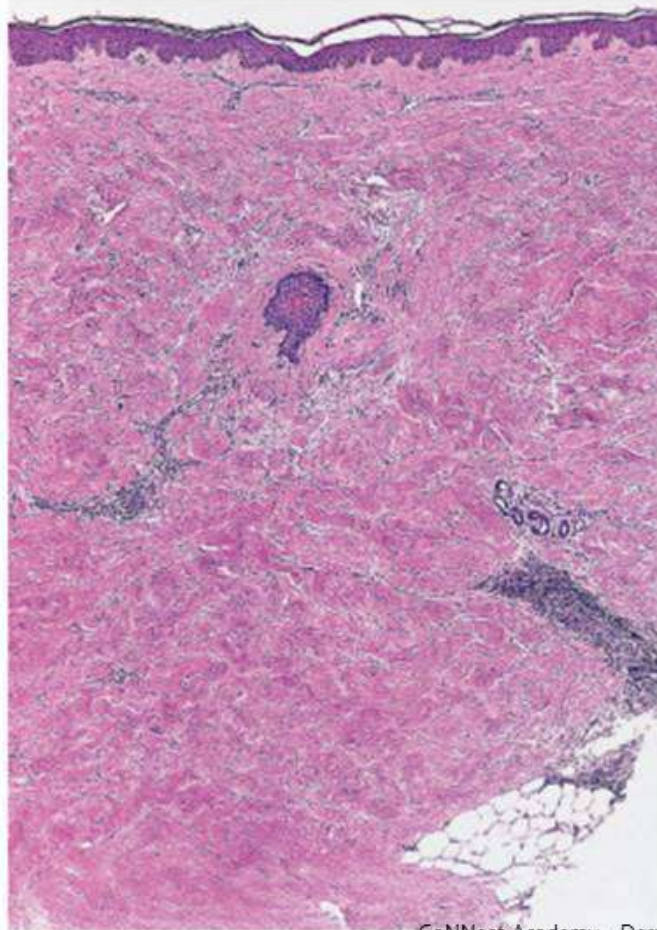


Fig. 43.12 Systemic sclerosis – histologic features. There is a dense sclerosis of the dermis with decreased adnexal structures and “trapping” of remaining adnexal structures that are encased by collagen. Sparse perivascular infiltrates of lymphocytes are present. *Courtesy, Ronald P Rapini, MD.*

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• Histopathology

- Excessive deposition of compact or hyalinized (homogeneous) collagen.
- A sparse lymphocytic infiltrate in the dermis
- Adnexal structures, especially eccrine sweat glands:
 - markedly atrophic.
 - surrounded with few or no adipocytes.
 - appear surrounded “trapped” by the excessive deposition of new collagen.
- - lie higher in the dermis due to replacement of most of the subcutaneous fat by newly formed collagen. ____

- Serology ???

- **Diagnostic criteria (American College of Rheumatology)**

- Major criterion: Symmetric cutaneous sclerosis proximal to the metacarpophalangeal (MCP) or metatarsophalangeal (MTP) joints
- Minor criteria:
 - o Sclerodactyly
 - o Digital pitted scars or loss of substance from finger pads
 - o Bi-basilar pulmonary fibrosis

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Diagnosis of SSc requires one major criterion OR two or more minor criteria

Treatment

Treatment of cutaneous sclerosis;

Unfortunately, most of the available therapies are ineffective.

- D-penicillamine
- Minocycline
- Methotrexate (if overlap polymyositis)
- Interferons
- Photopheresis
- Phototherapy (PUVA and UVA1 [340-400 nm])

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CUTANEOUS FEATURES OF SYSTEMIC SCLEROSIS – POSSIBLE TREATMENTS	
Cutaneous feature	Possible treatment options
Cutaneous sclerosis	• Topical therapies (see Ch. 44) • UV phototherapy (see Ch. 44) • D-penicillamine* • Minocycline** • Methotrexate (if no ILD)^{†,‡} • Mycophenolate mofetil (when ILD)
Raynaud phenomenon (Figs 43.9 & 43.10)	First-line • Cold avoidance • Hand and feet warming packets • Discontinue all tobacco products Second-line • Calcium channel blockers (e.g. nifedipine SR 30 mg daily-BID, amlodipine 2.5–10 mg daily) Third-line (especially when accompanied by ulceration) • Phosphodiesterase type 5 inhibitors (e.g. sildenafil, tadalafil) • α-Adrenergic blockers (e.g. prazosin) • Angiotensin II receptor blockers (e.g. losartan) • Endothelin receptor antagonists (e.g. bosentan) Fourth-line • Serotonin reuptake inhibitors (e.g. fluoxetine)
Cutaneous ulcers (Fig. 43.11)	• Avoid excessive debridement • Moist, non-adherent dressings • Therapies listed above for Raynaud phenomenon • Low-dose aspirin or clopidogrel • IV prostanoid (e.g. epoprostenol)
Calcinosis cutis (Fig. 43.8)	• Low-dose warfarin • Calcium channel blockers • Sodium thiosulfate
Mal telangiectasiae	• Lasers appropriate for vascular lesions

DIFFERENTIAL DIAGNOSIS OF SCLERODERMOID CONDITIONS

Disorder	Clinical features
Mucinoses	
• Scleredema	Induration of the upper back, neck and face; occasional internal involvement (see Ch. 46)
• Scleromyxedema	Waxy papules (often in a linear array); diffuse induration favoring the face, upper trunk, arms and thighs; monoclonal gammopathy; neurologic, gastrointestinal and pulmonary involvement (see Ch. 46)
Immunologic	
• Chronic scleroderma-like GVHD*	Variable sclerosis that often resembles plaque-type or deep morphea initially and then may become generalized; eosinophilic fasciitis (see Ch. 52)
• Eosinophilic fasciitis	Symmetric induration with a "pseudo-cellulite" appearance on the extremities (sparing hands and feet) (see text)
• Generalized morphea*	Expansion and coalescence of morphea plaques to involve a large portion of the trunk and extremities (see Ch. 44)
• Fibroblastic rheumatism	Sclerodactyly; fibrotic nodules on the hands
• Overlap syndromes	Sclerodactyly; myositis; arthritis
Paraneoplastic	
• POEMS syndrome	Sclerotic skin on the extremities (see Ch. 114)
• Amyloidosis (primary systemic) [†]	Diffuse induration favoring the face, distal extremities and trunk (see Ch. 47)
• Carcinoid syndrome	Sclerotic skin on the legs (see Table 53.2)
• Paraneoplastic scleroderma-like syndrome	Similar to systemic sclerosis, but older age at onset; seen in association with a monoclonal gammopathy, hairy cell leukemia, and solid tumors (e.g. breast, lung)
Neoplastic	
• Carcinoma <i>en cuirasse</i> *	Sclerodermoid encasement of the chest by metastatic carcinoma (usually breast cancer) (see Ch. 122)

*Can overlap with Scleroderma-like syndrome, which by definition is not

[†]Primary cutaneous amyloidosis can also occur in patients with systemic sclerosis and generalized morphea.

DIFFERENTIAL DIAGNOSIS OF SCLERODERMOID CONDITIONS	
Disorder	Clinical features
Metabolic	
• Diabetic cheiroarthropathy	Thickened skin and limited mobility of the hands (see Table 53.4)
• Porphyria cutanea tarda ^{a,†}	Morpheaform plaques in sun-exposed areas (see Ch. 49)
Neurologic	
• Reflex sympathetic dystrophy*	Painful, cold, swollen extremity eventually develops cutaneous sclerosis (see Ch. 6)
• Spinal cord injury	Sclerotic skin in affected areas
Toxin-mediated	
• Nephrogenic systemic fibrosis*	Associated with exposure to gadolinium-based contrast agents (new cases becoming rare) (see text)
• Eosinophilia-myalgia syndrome	Associated with L-tryptophan ingestion (US, 1989) (see text)
• Toxic oil syndrome*	Associated with toxic oil ingestion (Spain, 1981) (see text)
• Silicosis	Associated with occupations such as mining (see text)
Drug- or chemical-induced (see text)	
• Bleomycin*	Acrosclerosis; Raynaud phenomenon; pulmonary fibrosis (more common, usually no concurrent skin lesions)
• Taxanes*	Edema followed by sclerosis of the lower extremities; acrosclerosis; rarely, pulmonary fibrosis
• Vinyl chloride, chlorinated hydrocarbons*	Acrosclerosis; acral fibrotic papulonodules; Raynaud phenomenon; acro-osteolysis; pulmonary fibrosis
Venous insufficiency	
• Lipodermatosclerosis*	Woody induration and red-brown discoloration of the lower legs in the chronic phase; may also involve the pannus (see Ch. 100)
Genetic disorders	
• Restrictive dermopathy [§]	Tight, thin skin over the entire body; joint contractures; <i>LMNA</i> or <i>ZMPSTE24</i> mutations
• Hutchinson-Gilford progeria	Sclerotic skin on the lower trunk, buttocks and thighs; <i>LMNA</i> mutations (see Ch. 63)
• Werner syndrome	Tight, sclerotic skin on the distal extremities; <i>RECQL2</i> mutations (see Ch. 63)
• Stiff skin syndrome*	Fibrosis of the skin/fascia of the buttocks and thighs with hip contractures; <i>FBN1</i> mutations (see text)
• Phenylketonuria	Sclerotic skin on the thighs and buttocks with hip contractures; <i>PAH</i> mutations (see Ch. 63)
• MONA (multicentric osteolysis, nodulosis, and arthropathy)	Areas of skin thickening with overlying hyperpigmentation and hypertrichosis; subcutaneous nodules favoring hands/feet; digital swelling; <i>MMP2</i> mutations
• Weill-Marchesani syndrome 2	Sclerosis of the back and extensor surfaces of the extremities, with sparing of the face; also referred to as GEMSS (glaucoma, lens ectopia, microspherophakia, stiffness of the joints, and shortness) syndrome; <i>FBN1</i> mutations
• Reticular atrophy of the skin	Thinning of the skin of the distal extremities and face, with furrowing of the forehead; seborrhea; digital clubbing; <i>HPGD</i> mutations
• Ataxia-telangiectasia	Tight, sclerotic facial skin; <i>ATM</i> mutations (see Ch. 60)
• Unilateral scleroderma	Colorless to atrophic skin on dorsal surface of hand and foot; calcinosis; lentiginosis (see Ch. 63)

Morphea

- Inflammatory skin disease, primarily affecting the dermis and may extend to subcutaneous fat that evolves into scar formation (scar-like sclerosis).
 - There is NO involvement of internal organs



	Morphea	Systemic sclerosis
Clinical features	Sclerosis is patchy or linear <i>asymmetric</i> in distribution	Sclerosis is <i>symmetric</i> (starts at the hands and fingers then extends progressively proximal)
	NO other cutaneous features (other than sclerosis)	Presence of <i>other cutaneous features (other than sclerosis)</i> e.g. Raynaud's phenomenon, ...
	NO involvement of internal organs	There is involvement of internal organs (GIT, lung and others)
Histopathology	Both conditions are characterized by excessive collagen deposition and it may be difficult to be distinguished BUT morphea has a more intense inflammatory infiltrate than SSc in its earlier stages	
Serology	NOT a prominent feature of morphea, except for generalized and linear morphea (antibodies to ssDNA and histones)	ANA, Anti-Scl-70 (generalized) and anti-centromere (localized)

Plaque type	Most common , asymmetric , multiple on trunk 3 variant superficial, nodular, guttat
Linear	Most common in children (3types) 1. Linear morphea trunk , limp 2. Encoup desabre frontoparital region 3. Parry Romberg
Deep	Deep dermis, sc fat >>> tethered , bound down Disabling pan sclerotic morphea May affect muscle, bone Scc
Bullous	}2 anatomical site , }3 cm , } 4 plaques
generalized	



Fig. 44.3 Early inflammatory plaque-type morphea of the trunk. Early-stage lesion presenting as an erythematous edematous plaque.





Fig. 44.6 Nodular (keloidal) morphea. Elevated, firm pink papulonodules arising within an area of hyperpigmented induration. *Courtesy, Jean L. Bolognia, MD.*





Fig. 14-6 Linear morphea of an extremity. A Linear sclerotic band of the arm with both hyper- and hypopigmentation. The majority of patients with linear morphea have unilateral involvement. **B** More inflammatory phase with ulceration in addition to induration. The differential diagnosis includes linear melorheostosis which is often associated with underlying candlewax-like linear hyperostosis. A. Courtesy, Julie V Schaffer, MD.



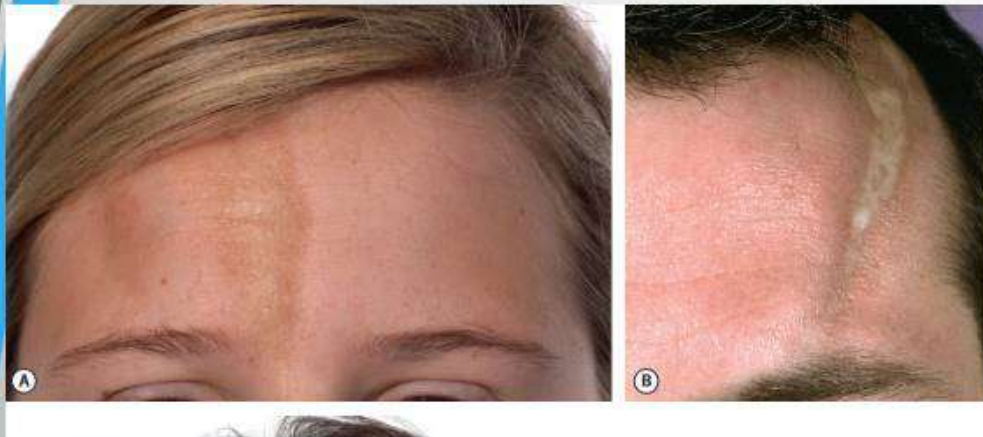


Fig. 44.7 Parry-Romberg syndrome. Unilateral sclerosis and loss of subcutaneous fat, leading to facial asymmetry. *Courtesy, Lorenzo Cerroni, MD.*



Fig. 44.11 Extensive linear morphea of the right leg leading to hypoplasia. In addition to marked hypoplasia, there is induration and a flexion contracture of the knee.



Fig. 44.10 Generalized morphea. **A** Multiple hyperpigmented plaques on the trunk that are becoming confluent. **B** Circumferential involvement with sparing of the central portion of the breast is a classic distribution pattern. *B, Courtesy, Luis Requena, MD.*

- all ages are affected with the peak incidence occurring between 20 and 40 years of age. Prevalence of morphea increases with age

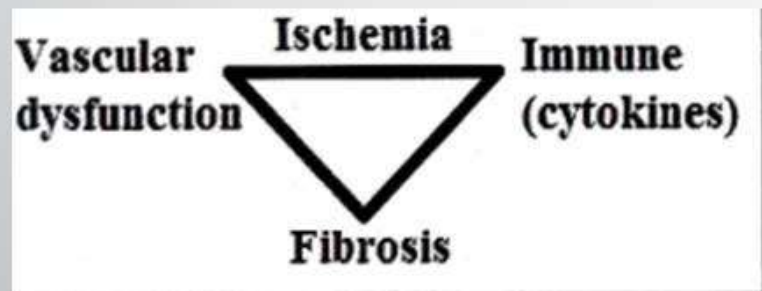
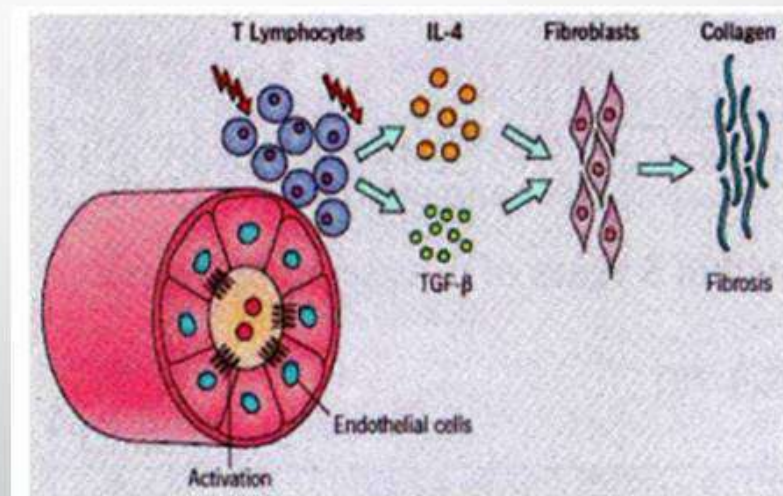
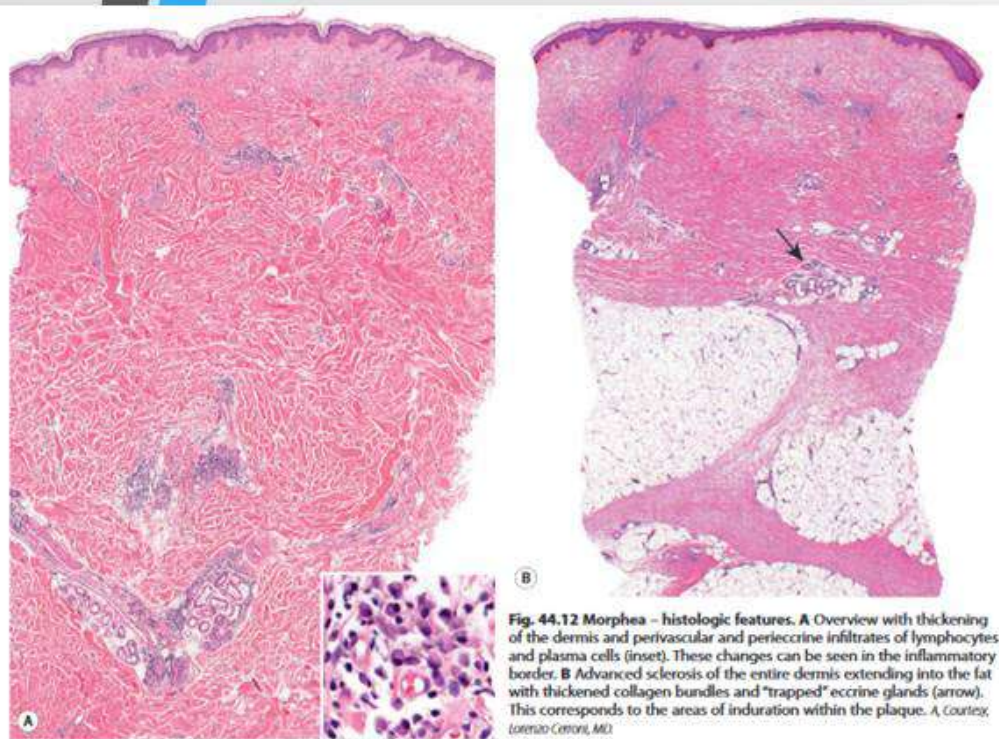


Fig 37.2 The three components involved in





- At the inflammatory border: endothelial swelling, edema and peri-vascular inflammatory infiltrate.

- In later stages:

- The epidermis is basically normal in appearance, but the rete ridges may be diminished, leaving a flattened dermal-epidermal junction (i.e. atrophic).

- Homogeneous collagen bundles with decreased space between the bundles

- Eccrine glands appear atrophic, “trapped” within the thickened dermis with decreased adipocytes around them and higher up in the dermis than usual because of increased collagen in subcutaneous fat

- Serology???

- **(A) Topical / Intra-lesional therapies:**
 - Indications: localized plaque morphea
- • Calcineurin inhibitors:
 - Corticosteroids e.g. clobetasol propionate
 - Vitamin D derivatives e.g. calcipotriol twice daily
 - Imiquimod: thrice weekly decreases erythema and induration of lesions
(mechanism of action: increased production of interferon-gamma).
 - Targeted (lesion-limited) phototherapy
 - Intralesional corticosteroids / interferon-γ

-) **Systemic therapies / Phototherapy:**
 - Indications:
 - Generalized morphea / Linear morphea involving the face or crossing joints
 - Immunosuppressive agents
- Corticosteroids: (methylprednisolone or prednisone 1-2 mg/kg/day) d
- Methotrexate: 15-25 mg for adults (0.3 mg / kg for children) / week
- Others:
 - Mycophenolate mofetil (MMF): inhibits type 1 collagen expression, enhances the expression of matrix metalloproteinase-1, alters fibroblast migratory and contractile functions, down regulates profibrotic cytokines TGF β i, and inhibits smooth muscle cell proliferation in arterials (these combined effects inhibit fibrosis)

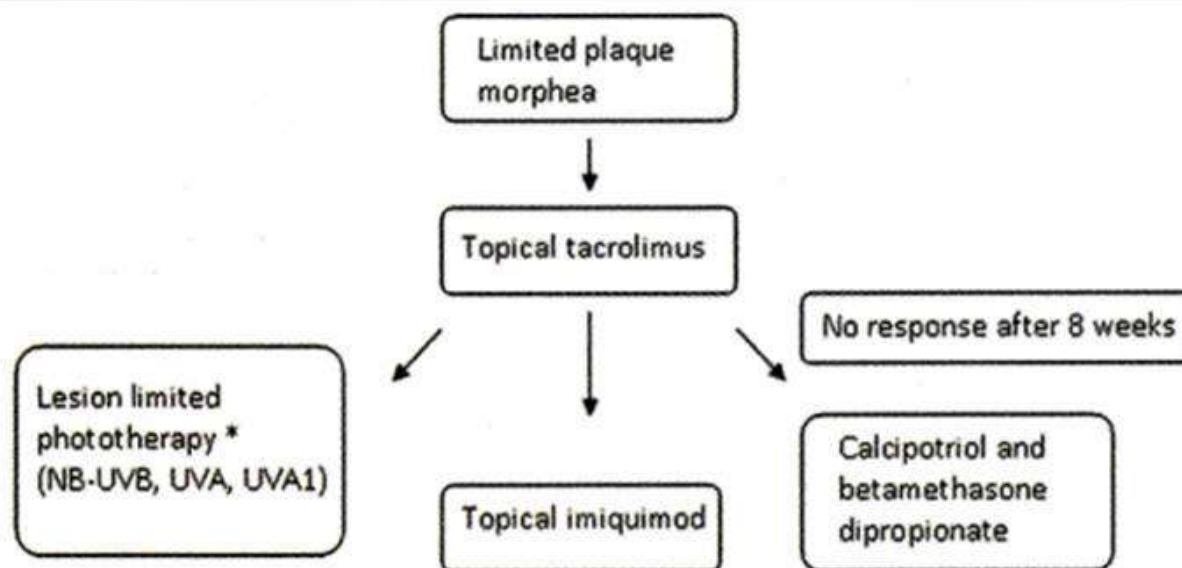


Fig 39.6 Treatment algorithm for limited plaque morphea.

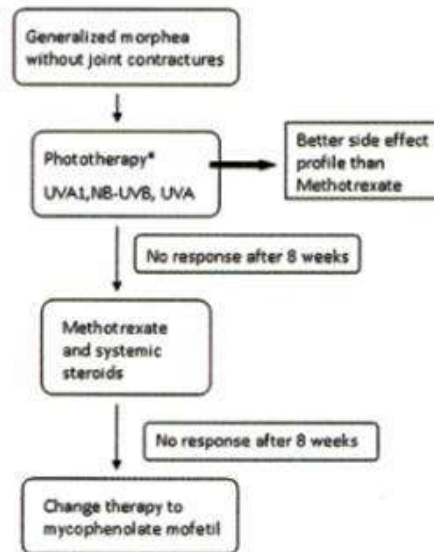


Fig 39.7 Treatment algorithm for generalized morphea

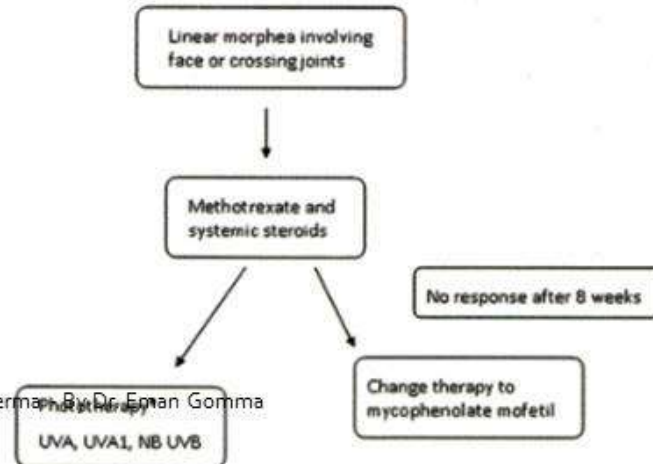


Fig 39.7 Treatment algorithm for linear morphea involving face or crossing joints.

Lichen sclerosus et atrophicus



Fig. 44.13 Lichen sclerosus. Shiny, ivory-colored plaques of the breast (A) and the lower back (B). Several guttate satellite lesions are present on the breast.



Fig. 44.16 Lichen sclerosus of the penis (balanitis xerotica obliterans). Note the ivory color, erosion and scarring.

Lichen sclerosus et atrophicus

- Sex; more common in females (10:1) -
 - Age: depends on site and sex
 - Genital LSA:
 - Females: 2 peaks (women in 5th- 6th decades, girls 8 - 13 years)
 - Males: all ages (boys and men)
 - Extra-genital LSA: rare in children

- lichen sclerosus et atrophicus (LSA) is an inflammatory disease
 - characterized by scar-like fibrosis and atrophy
 - primarily affecting the genital mucosa (male / female) - less often it affects the extra-genital skin
 - without systemic (internal organ) affection

Enviromental
triggers

Genaticly

Immune

dysregulation

susptable pt

- 1) Genetic predisposition;
 - Association with the MHC class II antigen HLA-DQ7.
 - Coincidence in monozygotic and dizygotic twins.
- (2) Triggers:
 - Oxidative stress may play a role in the pathogenesis
- Immunological dysregulation;
 - IgG autoantibodies against **extracellular matrix protein 1 (ECM-1)** are present in 80% of patients.

(1) Anogenital region.

	Females (Women)	Males (Boys and men)
Nomenclature	Kraurosis vulvae	Balanitis xerotica obliterans
Sites	Vulva and perianal region (figure-of-eight)	Glans and inner aspect of the foreskin (rarely perianal)
Symptoms	Severe pruritus (itching) and soreness (may be asymptomatic)	
Lesion	Shiny hypopigmented and sclerotic atrophic plaques 	Atrophic white sclerotic scar 
Complications	• Dysuria, dyspareunia, pain upon defecation (presents as constipation in children). • Scarring → burying of the clitoris and fusion of the labia minora to the labia majora • Dyspareunia (difficult or impossible sexual intercourse) due to significant narrowing of the introitus (severe cases) • Pre-cancerous (see below)	• Pain on erection (due to constrictions) • Dysuria and urinary obstruction (advanced stages) • If the foreskin is affected → inevitable "acquired" phimosis and paraphimosis = urologic emergency • Recurrent balanitis • Progressive disease → poorly healing ulcers of the glans • Pre-cancerous (see below)



Fig. 44.15 Anogenital lichen sclerosis. A Characteristic vulvar, perineal and perianal involvement in a child. Symmetric ivory-colored plaques with a thin pink peripheral rim are seen; a perianal pyramidal protrusion is also present. **B** Perianal lichen sclerosis in which the hypopigmented plaques are accompanied by a linear erosion (skin tear) at 6 o'clock and central fissuring of the perineum. A, Courtesy, A Hernández-Martín, MD.



Fig. 44.16 Lichen sclerosis of the penis (balanitis xerotica obliterans). Note the ivory color, erosion and scarring.



Fig. 44.13 Lichen sclerosus. Shiny, ivory-colored plaques of the breast (A) and the lower back (B). Several guttate satellite lesions are present on the breast.

- Extra-genital skin:
- Symptoms: frequently asymptomatic (except dryness / associated pruritus).
- Sites: trunk and the proximal extremities. neck, shoulders, flexor surfaces of the wrists, and sites of physical trauma or continuous pressure (e.g. shoulder or hip).
- Lesions:
 - Sclerotic, scar-like papules and plaques, ivory in color with a shiny surface
 - More advanced cases show telangiectasias or follicular plugging.
 - May be complicated by the development of bullae - usually hemorrhagic (due to the fragility of the dermal-epidermal junction)



Fig. 44.14 Lichen sclerosis. **A** Follicular plugging in a plaque of lichen sclerosis on the back of a patient with chronic GVHD. **B** Hemorrhagic bullae primarily at the periphery of a large plaque on the back. *A, Courtesy, Jean L. Bolognia, MD; B, Courtesy, Luis Requena, MD.*



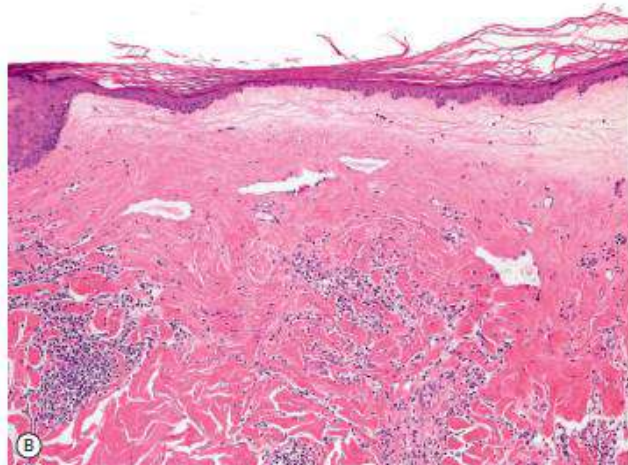
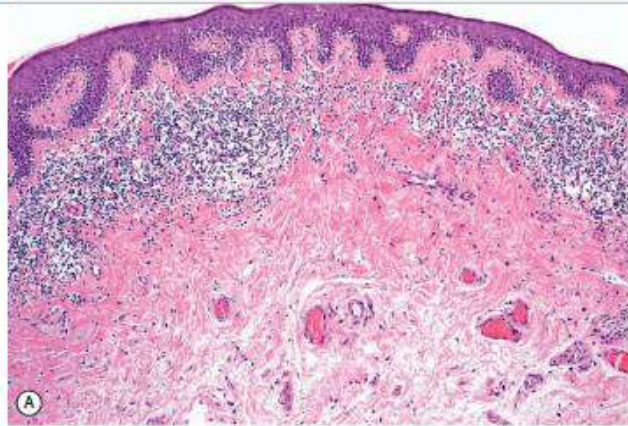


Fig. 44.17 Lichen sclerosis – histologic features. **A** Early lesion with band-like infiltrate of lymphocytes in the upper dermis and vacuolar change at the dermal–epidermal junction. Mild homogenization of the papillary dermis is present. **B** Late lesion with orthohyperkeratosis, thinning of the epidermis and homogenization of the upper dermis. There is also hyalinization and sclerosis of the papillary dermis, telangiectasias and an interstitial lymphocytic infiltrate.

Courtesy, Lorenzo Cerroni, MD.

- histopathology
LSA is a lichenoid interface dermatitis
- . Early: Superficial dermal edema (pale staining) + band-like lymphocytic infiltrate in the upper dermis
- - Late: (well-developed lesions)
Epidermis:
 - Thinned (atrophied)
 - Orthohyperkeratosis especially pronounced at follicular openings and may lead to follicular plugging
 - Vacuolar degeneration of the basal layer
 - Flattening of the rete ridges
- Superficial dermis:
 - Homogenized collagen + loss of elastic

	Morphea	L.SA
Epidermis	Relatively normal	Thinning (Atrophied)
	No follicular plugging	Follicular plugging
Dermal-epidermal junction	No hydropic degeneration	Hydropic degeneration
	Uncommon sub-epidermal	Often subepidermal bullae
	Whole dermis affected	Superficial dermis affected
Dermis	Appears homogenized	Marked edema
	Elastic fibers present	Elastic fibers absent
Subcutis	Inflammation / Fibrosis	No inflammation or fibrosis

TREATMENT OF MORPHEA AND LICHEN SCLEROSUS Rx					
		Morphea		Lichen sclerosis	
	Treatment modalities	Efficacy	Level of evidence	Efficacy	Level of evidence
Local	Topical corticosteroids	+	3	+++ (ultrapotent)	1
	Intralesional corticosteroids	+	3	++	2
	Topical calcineurin inhibitors	+ (early lesions)	2	++	1
	Imiquimod	+	2	No experience	
	Vitamin A analogues	+	3	+	2
	Vitamin D analogues	+	3	+	3
	Testosterone	No experience		0	1
	Progesterone	No experience		0	1
	Intralesional interferon- γ	0	1	No experience	
Systemic	Penicillin	++ (~5% of patients)	3	No experience	
	Hydroxy-/chloroquine	No experience		+	3
	Corticosteroids	+	3	+	3
	Vitamin A analogues*	+	3	+	1
	Vitamin D analogues	0	1	+	3
	Cyclosporine	0	3	No experience	
	Penicillamine	++	3	No experience	
	Methotrexate	++	2	No experience	
	Methotrexate plus pulsed corticosteroids	+++	2	+	2
Phototherapy	Oral photochemotherapy	++	3	+	3
	Bath photochemotherapy	+++	2	++	3
	Cream photochemotherapy	+	3	+	3
	UVA1	+++	2	++	2
	Photodynamic therapy	0 - +	3	+	3
	Extracorporeal photophoresis	+	3	No experience	
Others	CO ₂ laser	No experience		++	3
	Surgery	Selected patients		Selected patients	
	Physical therapy	Important		-	

*e.g. acitretin.

Table 44.2 Treatment of morphea and lichen sclerosis. Trials of ultrapotent topical corticosteroids have been performed primarily in women with vulvar lichen sclerosis. +++, Highly effective; ++, effective; +, moderately effective; 0, low efficacy or ineffective. Key to evidence-based support: (1) prospective controlled trial; (2) retrospective study or large case series; (3) small case series or individual case reports.

CONNECT Academy - Dermatology - By Dr. Eman Gomaa



THANK YOU!